S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Financial Risk Intelligence Platform

Real-time AI/NLP risk signals from news and social media, driving tactical index rebalancing and event-driven portfolio stress testing

Sentiment macro-F1 0.80

Both modules built

49 automated tests

Lakshay Gupta
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)
lakshay24318@iiitd.ac.in

Risk moves at headline speed; risk reports don't

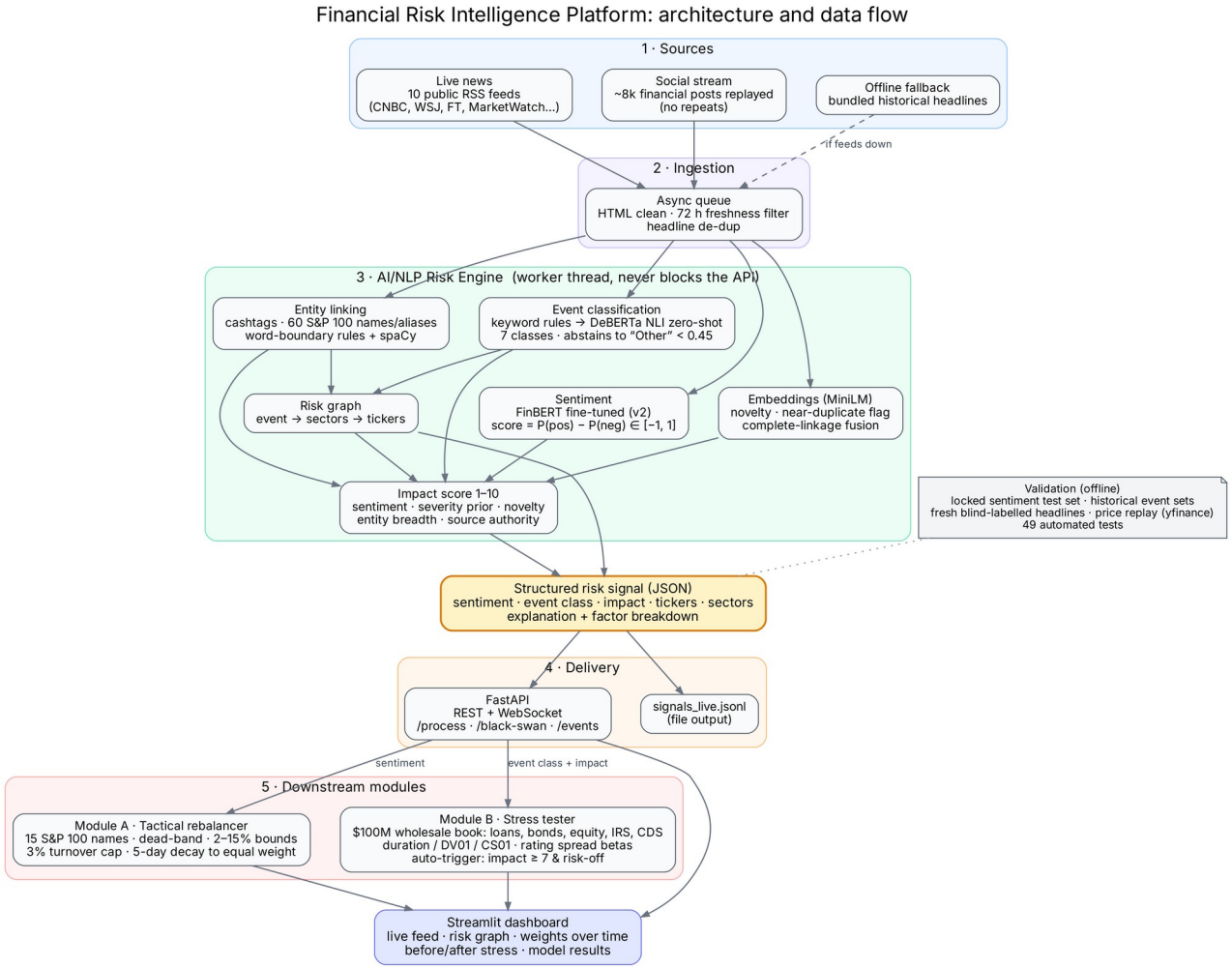
The problem

- Market-moving events arrive as unstructured text: news, filings, social posts
- Analysts cannot read everything, and keyword alerts are noisy
- Stress tests run on a quarterly calendar, not when the event happens
- **Goal: turn text into structured, explainable risk signals and act on them in seconds**

The approach

1. **Ingest**
   10 live RSS feeds + social stream; freshness filter and de-dup
2. **Understand**
   Fine-tuned FinBERT sentiment, event class (keywords → NLI), tickers
3. **Score**
   Explainable 1–10 impact: severity, novelty, breadth, source
4. **Act**
   Module A rebalances an index; Module B stress-tests a $100M book

System design: one engine, two consumers



Non-blocking

Models run in a worker thread with an inference lock; the API and dashboard stay responsive

Two outputs

Every signal goes to REST + WebSocket and to a JSONL file, as the brief allows either

Fails safe

No feeds? Bundled headlines replay. Fine-tuned model missing? Falls back to base FinBERT

Duplicates suppressed

Near-identical headlines (cosine ≥ 0.80) never move weights or trigger a second stress test

Implementation highlights and why

1

Fine-tuned FinBERT

Trained on 8,111 finance posts; locked test set never used for checkpoint selection; score = P(pos) − P(neg)

2

Classifier that abstains

Precise regex rules first, DeBERTa NLI for the rest; below 0.45 confidence it says “Other” instead of guessing

3

Explainable impact

Five transparent factors with fixed weights; every signal shows its point-by-point breakdown

4

Module A controls

Dead-band, 2–15% bounds, 3% turnover cap, 5-day decay to equal weight, 5 bp costs

5

Module B realism

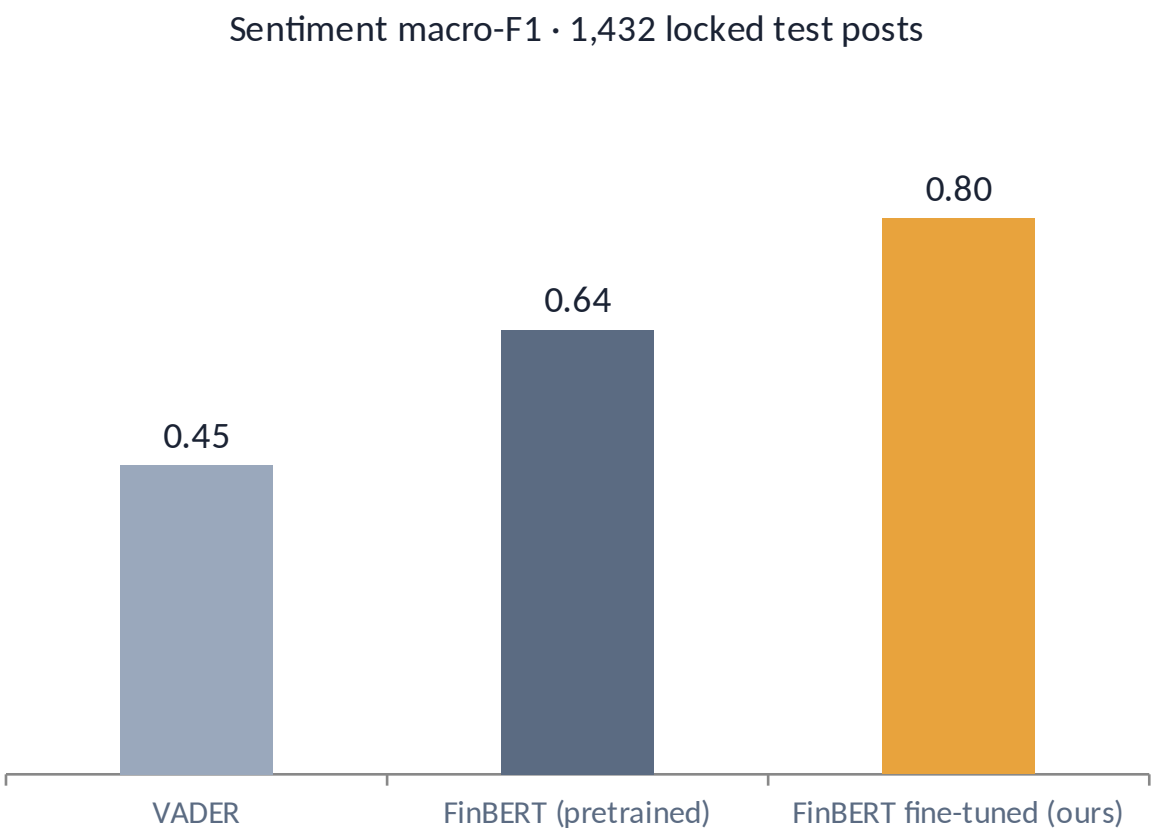
Duration, DV01/CS01, rating spread betas; rates hit the whole book; trigger needs impact ≥ 7 and risk-off

6

Engineering

49 offline tests incl. end-to-end API; pinned dependencies; one script regenerates every result

Results: +0.16 macro-F1 over FinBERT, 2× baseline on events



Accuracy 48.2% → 69.0% → 83.5% (majority class 64.7%)

Event classification accuracy vs always-guess-majority

64.9%
vs 32.4%
Fresh live headlines, labelled blind (n=37)

77.8%
vs 50.0%
Historical holdout (n=18)

74.1%
vs 51.9%
Historical tuning (n=27)

100% recall on Geopolitical and Credit Event (the classes that trigger stress tests) · ticker recall 75%

Domain impact: from headline to stress test in seconds

Live example

"China blockade of Taiwan Strait sparks military standoff; TSMC, NVIDIA and Apple shares plunge as sanctions loom"

Geopolitical · impact ≈ 8/10 · TSM, NVDA, AAPL
Stress test triggered automatically

≈ −6% of a $100M wholesale book: about $6M

Technology 64% of gross loss · bonds 53%
CDS protection offsets part of it · loss explained per position

1 **Early warning**

Every headline scored in about a second, with a factor breakdown an analyst can audit

2 **Event-driven stress testing**

Scenarios run when the event happens, not on a quarterly calendar; attributed by sector, asset and position

3 **Trustworthy by design**

Abstains when unsure, suppresses duplicates, never stresses on good news, keeps working offline

Market evidence, limits and next steps

What the market evidence shows

- Tweets (8,364 scored, 916 stock-days, 2017–18): weak but consistent contrarian signal, IC −0.06 to −0.09 over 1–3 days, negative in both halves; TextBlob agrees
- So a momentum tilt is the wrong sign for social data: Module A replay ties the benchmark (+0.11 pp)
- News headlines: direction 51% on 45 events, a coin flip at this n; impact score is triage, not a forecast
- Module B uses first-order sensitivities; social feed is a replay, not a live API

Next steps

1. **Test a contrarian tilt for social sources**
   On later, unseen data (2019+), with news signals kept separate
2. **Calibrate impact on a larger archive**
   Fit impact weights to thousands of dated headlines and realised moves
3. **Credit losses, not just mark-to-market**
   Add PD/LGD and rating migration to Module B; connect a live social API

Everything above is reproducible: bash scripts/reproduce_results.sh